

Orders and Q numbers

#####

Ok... I see a problem here. Q is the filament count. A filament is a fundamental particle. So, $Q=3$ is a quark, not He3. He3 has a Q of 9. But it's self-similar, so it may still approximate the mass of He3 because its nucleus is a higher-order of 3-braid... To break it down, He3: $Q=9$, 2nd order supercoil (eg three first order $Q=3$ nucleons wrapped as a second order $Q=9$ braid). For all atomic nuclei, coil order is 2 (probably) and Q is equal to atomic number times 3

So:

Vacuum: $Q=1$, coiling order = 0? n/a?

photon, neutrino: $Q=1$, coiling order = 1 (BUT these are single-coil helices... they ride along filaments, pushed by the time wave with minimal resistance... small normal force to time pressure as a result of filament resistance to kinking = 'pseudomass' ...they are defined as bosons or 'true bosons' in this model because they don't have extended coil in time)

electron, quark: $Q=1$, coiling order 1, coil type = persistent, extended helix in the 4th dimension, resists time pressure = mass)

mesons: $Q=2$, coiling order = 1, persistent braiding of two intercoiled quark filaments = strong force

nucleons: $Q=3$, coiling order = 1, persistent braiding of three intercoiled quark filaments = strong force, may experience mass suppression via first-order 'braid smoothing'

Atomic nuclei: 2nd order coils, the 2nd=order nucleonic braiding is treated as 'velcro action' between the coils of the quarks (as the quark triplets 'bend around the braid' their 1st order coils 'open up' to interaction, which allows nucleons to bind into nuclei (superposition is an equally valid interpretation, it's just the mechanical metaphor makes it intuitive), this 'opening up' also allows the internal coiling to do this 'velcro-style' meshing with the 1st-order coils of electrons, hence the electroweak force—both result from the meshing of 1st-order coils (and forces operating at higher orders of coiling are essentially the same thing, with some

adjustments due to scaling and geometric limitations—in other words, it's solenoids all the way down, just interpreted as springs/helices); electrons are not considered to be 'intercoiled' with nuclei (I don't think) they're sort of 'accessory coils' forming a bundle—although now that I think of it, certain geometries of electron shell would probably exhibit geometric mass suppression though these patterns would be more subtle for anything that's very far up the periodic table. The light nuclei would see the effect more.

H1 = Q=3, coiling order = 2, may experience mass suppression via second-order 'braid smoothing'

Atomic nuclei

He3 = Q=9, coiling order = 2, may experience mass suppression via second-order 'braid smoothing'

Atomic nuclei

Li4 = Q=12, coiling order = 2, geometrically unstable

Actually, I'm not sure if H1 would be just a first order proton, or if it's a second order coil of that proton filament bundle. Perhaps deuterium is a second-order coil of H1 in which case D wouldn't actually have a neutron it would just be an anomalously heavy hydrogen atom; **very anomalously heavy, so if 2nd-order hydrogen is a thing, I think it would likely appear as a strange 'other' particle... let me think this through...**

Either all hydrogen-1 nuclei are identical to a proton, in which case SAT should calculate everything for hydrogen exactly as a proton's 1st-order 3-quark braid.

Or all hydrogen-1 nuclei are 2nd-order coils of that 1st-order 3-quark braid, in which SAT should make a distinction between a proton and an H1 nucleus, but if that were true then science would already have discovered some weird (quite large) mass or size anomaly between the way atomic hydrogen behaves and the way bare protons or protons in nuclei behave. It would mean H1 would be a distinct particle from a proton.

Or a 2nd-order coiling of a proton is deuterium (but that would mean deuterium doesn't have a neutron), or there's some other particle that represents 'double-coiled hydrogen-1' that we've misinterpreted and classified as some other thing (but that thing would appear as a strongly large and maybe massive H1 and not be decomposable to something smaller other than quarks), or protons simply don't exhibit second-order coiling except when participating in a nucleus with other nucleons.

It would probably be very large and fairly stable, maybe *very* stable... probably massive but light for its size. Possibly with weird 4d morphology, which would manifest as unusual changes in properties. Possibly quite common.... But more likely only occurs under high energy or maybe extreme gravity conditions or doesn't occur at all

Q count is *always* identical to fundamental quark count, and it's *always* an integer. Although, there's some theoretical work we have to do on interpreting Q count for larger particles, because it's a fractal relationship, you can 'squint' and treat He3 a Q=3 system (where the triple-coiled quarks are coarse grained out of resolution and each nucleon is treated as a single filament) and the calculations will probably come close.